GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Fatty Acid Metabolic Process (GO:0006631	−0.21034135	108	4.808e−14	2.590e−10	CYP1A1:41 ABCB11:82 CPT1A:139 PLA2G1B:162 ABCD3:276 ACADSB:297
Fatty Acid Catabolic Process (GO:0009062	−0.26294838	58	4.529e−12	1.220e−08	EHHADH:5 ECI2:151 ABHD3:188 ABCD3:276 ACADM:312 NUDT7:363
Fatty Acid Beta−Oxidation (GO:0006635)	−0.27437488	47	7.889e−11	1.417e−07	EHHADH:5 GCDH:10 ECI2:151 ABCD3:276 ACADM:312 ACAD11:371
Monocarboxylic Acid Transport (GO:001571	−0.22901951	61	6.416e−10	8.641e−07	SLC5A12:3 SLC16A5:25 SLC10A2:30 SLC22A13:32 SLC6A12:36 FABP1:72
Organic Hydroxy Compound Transport (GO:0	−0.29637363	35	1.323e−09	1.426e−06	SLC10A2:30 SLC51A:215 ATP8B1:229 ABCD3:276 RALBP1:286 ABCG5:296
Fatty Acid Oxidation (GO:0019395)	−0.25441988	47	1.642e−09	1.474e−06	EHHADH:5 ECI2:151 ABCD3:276 ACADM:312 ACAD11:371 DECR1:380
Monocarboxylic Acid Metabolic Process (G	−0.18461497	80	1.196e−08	9.203e−06	AGXT:39 CYP1A1:41 ABCB11:82 HOGA1:103 NPC1:125 CPT1A:139
Nitrogen Compound Transport (GO:0071705)	−0.13734075	144	1.397e−08	9.406e−06	SLC22A13:32 SLC6A12:36 SLC6A6:40 RHAG:53 RHD:129 RHCE:143
Alpha−Amino Acid Catabolic Process (GO:1	−0.29766639	29	2.926e−08	1.588e−05	TAT:24 GOT2:93 HOGA1:103 MCCC1:204 SARDH:213 HIBCH:223
Steroid Metabolic Process (GO:0008202)	−0.20547412	61	2.948e−08	1.588e−05	CYP11A1:22 DHRS4:33 CYP1A1:41 CYP17A1:63 NPC1:125 NR1I2:158
Chemical Synaptic Transmission (GO:00072	0.10099280	237	9.769e−08	4.784e−05	SNAP25:11 GABRR1:56 NPTX1:57 GRIK3:102 SYT1:113 GRIK1:165
Modulation Of Chemical Synaptic Transmis	0.14917660	105	1.355e−07	6.082e−05	CPLX3:5 NPTX1:57 GRIK3:102 SYT1:113 LRRC4:163 GRIK1:165
Amino Acid Catabolic Process (GO:0009063	−0.23582970	38	4.967e−07	2.058e−04	GOT2:93 HOGA1:103 MCCC1:204 SARDH:213 HIBCH:223 BCKDK:767
Cell Morphogenesis Involved In Neuron Di	0.18118493	64	5.522e−07	2.125e−04	SLITRK4:70 SLITRK1:138 TRIM46:166 NRCAM:173 CNTN4:359 POU4F1:636
Anterograde Trans−Synaptic Signaling (GO	0.11126573	169	6.550e−07	2.352e−04	SNAP25:11 GABRR1:56 NPTX1:57 SYT1:113 GRIK1:165 CHRM2:196
Regulation Of Heart Rate By Cardiac Cond	0.23265996	36	1.378e−06	4.641e−04	KCND3:69 KCNH6:241 SCN3B:398 CAV1:644 BIN1:1043 KCNE3:1058
Nervous System Development (GO:0007399)	0.07405301	363	1.501e−06	4.756e−04	MAB21L2:12 DCLK1:55 NPTX1:57 MOG:71 VAX2:98 SLITRK1:138
Cardiac Conduction (GO:0061337)	0.21623555	41	1.687e−06	5.048e−04	KCND3:69 KCNH6:241 SCN3B:398 CAV1:644 BIN1:1043 KCNE3:1058
Bile Acid And Bile Salt Transport (GO:00	−0.34025941	16	2.461e−06	6.977e−04	SLC10A2:30 SLC51A:215 ATP8B1:229 ABCD3:276 SLCO1B1:300 SLC10A6:580
Neuron Projection Morphogenesis (GO:0048	0.11946541	124	4.568e−06	1.158e−03	EPHA4:41 DCLK1:55 SLITRK4:70 CDC42:77 SLITRK1:138 TRIM46:166
Sensory Perception Of Light Stimulus (GO	0.13930157	91	4.535e−06	1.158e−03	CRYAA:1 CRYBA1:36 CRYBB2:43 GNAT2:78 VAX2:98 TRPM1:111
Visual Perception (GO:0007601)	0.13980297	90	4.727e−06	1.158e−03	CRYAA:1 CRYBA1:36 CRYBB2:43 GNAT2:78 VAX2:98 TRPM1:111
Phosphatidylcholine Metabolic Process (G	−0.19321026	46	5.890e−06	1.379e−03	PNLIPRP2:46 PLA2G1B:162 ABHD3:188 SLC44A4:222 PLA2G2D:278 TSPO:757
Alpha−Amino Acid Biosynthetic Process (G	−0.41188060	10	6.489e−06	1.455e−03	SDS:47 BCAT2:114 CPS1:261 SDSL:383 ILVBL:1117 BCAT1:1461
Regulation Of Synaptic Transmission, Glu	0.18992748	47	6.754e−06	1.455e−03	GRIK3:102 SYT1:113 GRIK1:165 GRM2:199 GRM8:239 GRIN2B:294
Regulation Of Neuron Projection Developm	0.10400349	156	7.834e−06	1.623e−03	CAMK2B:4 ALKAL1:40 EPHA4:41 NCS1:48 NDEL1:148 ATP1B2:211
Quaternary Ammonium Group Transport (GO:	−0.36385737	12	1.278e−05	2.549e−03	SLC44A4:222 SLC22A4:263 SLC22A1:379 SLC6A20:576 SLC44A5:1051 SLC5A7:1554
Regulation Of Transcription By RNA Polym	0.03406669	1527	1.702e−05	3.274e−03	UBE2D1:15 UBE2I:19 HSF4:26 RPL17:38 CRTC3:54 ELOC:72
Axonogenesis (GO:0007409)	0.09722793	164	1.845e−05	3.314e−03	EPHA4:41 SLITRK4:70 SLITRK1:138 TRIM46:166 NRCAM:173 ULK2:301
One−Carbon Compound Transport (GO:001975	−0.22997048	29	1.831e−05	3.314e−03	SLC14A2:14 SLC4A1:17 RHAG:53 CA4:144 CA2:146 SLC4A7:1189
Fatty Acid Transport (GO:0015908)	−0.20759219	35	2.158e−05	3.695e−03	FABP1:72 GOT2:93 SLC22A1:379 SLC22A7:493 PPARD:866 SLC22A6:912
Sensory Perception Of Smell (GO:0007608)	−0.15239075	65	2.195e−05	3.695e−03	OR13C9:167 OMP:736 OR4D1:800 OR2Z1:847 OR6N2:925 OR5B21:1158
Axon Development (GO:0061564)	0.13349866	84	2.410e−05	3.873e−03	SLITRK4:70 SLITRK1:138 TRIM46:166 NRCAM:173 CNTN4:359 GPM6B:375
Ethanol Metabolic Process (GO:0006067)	−0.43085401	8	2.444e−05	3.873e−03	SULT1A2:132 SULT1A3:138 SULT1E1:170 SULT1B1:848 ACSS2:856 ADH4:1199
Carboxylic Acid Transport (GO:0046942)	−0.17401426	49	2.547e−05	3.920e−03	SLC5A12:3 SLC16A5:25 SLC6A12:36 SLC6A6:40 SLC6A19:169 SLC43A1:179
Glutamate Receptor Signaling Pathway (GO	0.21620676	31	3.122e−05	4.672e−03	GRIK3:102 TRPM1:111 HOMER2:133 GRIK1:165 GRM2:199 GRM8:239
Organic Cation Transport (GO:0015695)	−0.21388883	31	3.793e−05	5.523e−03	SLC6A12:36 SLC44A4:222 SLC22A4:263 RALBP1:286 TSPO2:370 SLC22A1:379
Branched−Chain Amino Acid Metabolic Proc	−0.30101820	15	5.441e−05	7.713e−03	BCAT2:114 MCCC1:204 HIBCH:223 BCKDK:767 PCCB:1050 IVD:1094
Lipid Transport (GO:0006869)	−0.12413848	88	5.852e−05	7.881e−03	EHHADH:5 SLC10A2:30 FABP1:72 ABCB11:82 GOT2:93 MTTP:123
Regulation Of Cell Differentiation (GO:0	0.09297735	158	5.795e−05	7.881e−03	FGF8:46 PHLDB1:67 TENT5C:143 DDX17:184 MTA3:216 MTA1:518

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
HSIAO_LIVER_SPECIFIC_GENES	−0.18866666	196	1.004e−19	6.511e−16	ALB:9 TAT:24 AGXT:39 SDS:47 ALDH1A1:56 FABP1:72
CARRILLOREIXACH_HEPATOBLASTOMA_VS_NORMAL	−0.08449157	911	1.162e−17	2.524e−14	EHHADH:5 HMCES:7 GCDH:10 CYP11A1:22 TAT:24 GNMT:29
REACTOME_METABOLISM_OF_LIPIDS	−0.09901782	651	1.167e−17	2.524e−14	EHHADH:5 ALB:9 CYP11A1:22 SLC10A2:30 CYP1A1:41 SPTLC3:43
REACTOME_FATTY_ACID_METABOLISM	−0.18539336	155	1.818e−15	2.948e−12	EHHADH:5 CYP1A1:41 ELOVL7:85 SCD:92 CPT1A:139 ECI2:151
REACTOME_OXIDATIVE_STRESS_INDUCED_SENESC	0.25351230	76	2.243e−14	2.909e−11	RPS27A:14 UBA52:20 H4C2:34 H4C14:45 H3C13:59 H4C15:75
REACTOME_BIOLOGICAL_OXIDATIONS	−0.17607463	146	2.248e−13	2.431e−10	MGST3:15 CYP11A1:22 CYP1A1:41 ALDH1A1:56 CYP2F1:57 CHAC2:66
REACTOME_NEURONAL_SYSTEM	0.11350397	352	3.144e−13	2.549e−10	CAMK2B:4 SNAP25:11 ADCY1:37 PRKCG:53 GABRR1:56 KCND3:69
REACTOME_OLFACTORY_SIGNALING_PATHWAY	−0.20619519	105	3.034e−13	2.549e−10	OR13C9:167 OR10V1:230 OR5AR1:245 OR52K2:421 OR4D11:731 OR4D1:800
REACTOME_DEVELOPMENTAL_BIOLOGY	0.07511183	792	1.056e−12	7.612e−10	RPL34:9 RPS27A:14 POLR2J:17 UBA52:20 MAGOH:21 H4C2:34
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	−0.13263648	232	3.842e−12	2.492e−09	SLC5A12:3 SLC14A2:14 SLC4A1:17 SLC6A12:36 SLC6A6:40 SLC13A2:50
REACTOME_CELLULAR_SENESCENCE	0.17228444	135	5.118e−12	3.018e−09	RPS27A:14 UBE2D1:15 UBA52:20 H4C2:34 H4C14:45 H3C13:59
REACTOME_NERVOUS_SYSTEM_DEVELOPMENT	0.09596927	436	8.025e−12	4.338e−09	RPL34:9 RPS27A:14 UBA52:20 MAGOH:21 RPL17:38 EPHA4:41
KEGG_VALINE_LEUCINE_AND_ISOLEUCINE_DEGRA	−0.29742758	43	1.518e−11	7.575e−09	EHHADH:5 ACAD8:104 BCAT2:114 MCCC1:204 HIBCH:223 ACADSB:297
REACTOME_CELLULAR_RESPONSES_TO_STIMULI	0.08031929	589	3.687e−11	1.708e−08	CAMK2B:4 RPL34:9 RPS27A:14 UBE2D1:15 UBA52:20 H4C2:34
REACTOME_NEUROTRANSMITTER_RECEPTORS_AND_	0.15162121	159	4.491e−11	1.821e−08	CAMK2B:4 ADCY1:37 PRKCG:53 GABRR1:56 GRIK3:102 NBEA:125
CAIRO_HEPATOBLASTOMA_DN	−0.12593522	232	4.336e−11	1.821e−08	GCDH:10 GNMT:29 AGXT:39 SDS:47 TMPRSS2:51 SLC1A1:60
REACTOME_METABOLISM_OF_AMINO_ACIDS_AND_D	−0.11050687	299	5.689e−11	1.942e−08	GCDH:10 TAT:24 GNMT:29 SLC6A12:36 AGXT:39 SDS:47
CHIANG_LIVER_CANCER_SUBCLASS_PROLIFERATI	−0.15486546	151	5.446e−11	1.942e−08	EHHADH:5 GCDH:10 TAT:24 AGXT:39 ABCB11:82 DHTKD1:98
KEGG_OLFACTORY_TRANSDUCTION	−0.17418552	119	5.547e−11	1.942e−08	OR13C9:167 OR10V1:230 OR5AR1:245 CLCA1:377 OR52K2:421 CALM2:502
BENPORATH_ES_WITH_H3K27ME3	0.06390560	913	9.263e−11	3.005e−08	BNC1:2 CAMK2B:4 MAB21L2:12 HSF4:26 EGFLAM:35 ALKAL1:40
WP_MALIGNANT_PLEURAL_MESOTHELIOMA	0.09759329	369	1.434e−10	4.431e−08	ACTC1:28 FGF8:46 ACTA2:85 ITGB4:91 EED:109 ACTG2:130
REACTOME_INTRACELLULAR_SIGNALING_BY_SECO	0.11758207	250	1.712e−10	5.047e−08	CAMK2B:4 RPS27A:14 UBA52:20 ADCY1:37 FGF8:46 PRKCG:53
WOO_LIVER_CANCER_RECURRENCE_DN	−0.22251131	66	4.163e−10	1.174e−07	GCDH:10 ACY1:83 GOT2:93 MTTP:123 HGD:124 SLC2A9:202
BENPORATH_SUZ12_TARGETS	0.06269669	852	7.580e−10	2.049e−07	BNC1:2 MAB21L2:12 HSF4:26 EGFLAM:35 ALKAL1:40 EPHA4:41
REACTOME_SIGNALING_BY_RECEPTOR_TYROSINE_	0.08609907	436	8.525e−10	2.212e−07	RPS27A:14 POLR2J:17 UBA52:20 FGF8:46 RHOA:52 MAPKAP1:73
LEE_LIVER_CANCER_DENA_DN	−0.21816056	66	9.033e−10	2.254e−07	EHHADH:5 PNLIP:8 GCDH:10 CTRC:13 ALDH1A1:56 CYP2F1:57
KIM_ALL_DISORDERS_OLIGODENDROCYTE_NUMBER	0.07335633	603	9.904e−10	2.295e−07	CAMK2B:4 UBE2M:13 RPS27A:14 POLR2J:17 RACK1:39 RAN:44
KIM_ALL_DISORDERS_CALB1_CORR_UP	0.08522208	442	9.796e−10	2.295e−07	CAMK2B:4 SNAP25:11 VSNL1:31 RAN:44 DCLK1:55 NPTX1:57
KIM_BIPOLAR_DISORDER_OLIGODENDROCYTE_DEN	0.07726247	531	1.402e−09	3.137e−07	CAMK2B:4 RPS27A:14 POLR2J:17 VSNL1:31 RAN:44 RHOA:52
BENPORATH_EED_TARGETS	0.06044436	860	2.546e−09	5.505e−07	BNC1:2 CAMK2B:4 MAB21L2:12 HSF4:26 TM6SF1:30 H4C2:34
LIM_MAMMARY_STEM_CELL_UP	0.08328880	433	3.324e−09	6.955e−07	BNC1:2 VSNL1:31 PHLDB1:67 ACTA2:85 PLPP1:88 ITGB4:91
REACTOME_MAPK_FAMILY_SIGNALING_CASCADES	0.10476050	264	5.120e−09	1.038e−06	CAMK2B:4 RPS27A:14 UBA52:20 FGF8:46 ICMT:50 CDC42:77
HOSHIDA_LIVER_CANCER_SUBCLASS_S3	−0.11037430	235	6.092e−09	1.198e−06	EHHADH:5 SLC23A1:6 SLC6A12:36 AGXT:39 SDS:47 ALDH1A1:56
KEGG_GLYCINE_SERINE_AND_THREONINE_METABO	−0.30131013	31	6.431e−09	1.227e−06	GNMT:29 AGXT:39 SDS:47 SARDH:213 BHMT:225 DMGDH:232
HOUNKPE_HOUSEKEEPING_GENES	0.05879803	860	6.784e−09	1.255e−06	SRSF3:7 SNRPD3:10 POLR2J:17 SF3B6:18 UBE2I:19 MAGOH:21
REACTOME_SIGNALING_BY_WNT	0.11675730	208	6.967e−09	1.255e−06	RPS27A:14 UBA52:20 H4C2:34 H4C14:45 RHOA:52 PRKCG:53
NIKOLSKY_BREAST_CANCER_8P12_P11_AMPLICON	−0.23607891	50	7.800e−09	1.358e−06	ANK1:16 ADAM2:21 IDO2:91 NSD3:118 ADAM9:119 CHRNA6:178
KEGG_FATTY_ACID_METABOLISM	−0.27792881	36	7.956e−09	1.358e−06	EHHADH:5 GCDH:10 CPT1A:139 ECI2:151 ACADSB:297 ACADM:312
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	0.15168660	120	9.899e−09	1.647e−06	DNAJC5:151 ARFRP1:167 LAMA5:263 ADRM1:319 CHRNA4:339 CBLN4:358
REACTOME_ESR_MEDIATED_SIGNALING	0.14476716	131	1.103e−08	1.789e−06	POLR2J:17 H4C2:34 H4C14:45 H3C13:59 H4C15:75 H2BC15:97

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Hyperammonemia	−0.17859367	117	2.756e−11	2.702e−07	MANBA:55 NPC1:125 NAGLU:131 CPT1A:139 MCCC1:204 NAGS:211
Drug−Induced Liver Disease	−0.11552404	200	1.999e−08	9.798e−05	ALB:9 CYP1A1:41 IL1R2:49 CA3:71 ABCB11:82 CRYZ:107
Cystic Fibrosis	−0.07723043	421	6.952e−08	2.272e−04	BPIFB1:2 ALB:9 CTRC:13 SDS:47 LCT:86 GOT2:93
Cholestasis	−0.09594453	247	2.348e−07	5.755e−04	SLC23A1:6 SLC10A2:30 CYP17A1:63 ABCB11:82 HOGA1:103 NPC1:125
Renal salt wasting	−0.37348283	15	5.515e−07	1.081e−03	CYP11A1:22 KCNJ1:37 CLCNKA:108 CLCNKB:134 SCNN1A:212 SCNN1B:217
Metabolic acidosis	−0.17751203	65	7.652e−07	1.250e−03	EHHADH:5 GCDH:10 SLC4A1:17 AGXT:39 LCT:86 SLC9A4:99
HYPERGLYCINURIA (disorder)	−0.36616279	15	9.139e−07	1.280e−03	SLC6A19:169 ACADM:312 SLC6A20:576 GCSH:921 PCCB:1050 ALDH4A1:1062
Kidney Failure, Chronic	−0.07624834	345	1.331e−06	1.290e−03	ALB:9 SLC4A1:17 KCNJ1:37 AGXT:39 CYP1A1:41 FABP1:72
Hepatomegaly	−0.07879913	325	1.210e−06	1.290e−03	EHHADH:5 GCDH:10 GNMT:29 CYP1A1:41 RHAG:53 ABCB11:82
Steatohepatitis	−0.06626893	458	1.447e−06	1.290e−03	ALB:9 SLC4A1:17 CA3:71 FABP1:72 SCD:92 MTTP:123
Vomiting	−0.10946579	167	1.138e−06	1.290e−03	GCDH:10 CYP11A1:22 KCNJ1:37 COMT:74 XPC:101 HTR3C:141
Glycinuria	−0.38189414	13	1.869e−06	1.527e−03	SLC6A19:169 ACADM:312 SLC6A20:576 GCSH:921 PCCB:1050 ALDH4A1:1062
Fabry Disease	−0.21396582	41	2.161e−06	1.630e−03	MANBA:55 CHDH:241 SI:267 VDR:305 ABCB1:335 GLA:651
Autism Spectrum Disorders	0.06469175	456	2.681e−06	1.878e−03	SNAP25:11 IQGAP3:16 ACTC1:28 JAKMIP1:32 ETF1:47 NCS1:48
Hepatitis, Toxic	−0.10528953	161	4.291e−06	1.913e−03	ALB:9 IL1R2:49 CA3:71 NR1I2:158 SULT1E1:170 CAT:240
Chemical and Drug Induced Liver Injury	−0.10528953	161	4.291e−06	1.913e−03	ALB:9 IL1R2:49 CA3:71 NR1I2:158 SULT1E1:170 CAT:240
Chemically−Induced Liver Toxicity	−0.10528953	161	4.291e−06	1.913e−03	ALB:9 IL1R2:49 CA3:71 NR1I2:158 SULT1E1:170 CAT:240
Drug−Induced Acute Liver Injury	−0.10528953	161	4.291e−06	1.913e−03	ALB:9 IL1R2:49 CA3:71 NR1I2:158 SULT1E1:170 CAT:240
Hyperactive renin−angiotensin system	−0.47123765	8	3.923e−06	1.913e−03	KCNJ1:37 CLCNKB:134 SLC26A3:203 SCNN1A:212 SCNN1B:217 SCNN1G:772
Inborn Errors of Metabolism	−0.16452708	66	3.887e−06	1.913e−03	GCDH:10 AGXT:39 ACY1:83 HEXB:94 HGD:124 CA2:146
Increased plasma renin activity	−0.47123765	8	3.923e−06	1.913e−03	KCNJ1:37 CLCNKB:134 SLC26A3:203 SCNN1A:212 SCNN1B:217 SCNN1G:772
Lysosomal Storage Diseases	−0.15733284	72	4.015e−06	1.913e−03	MANBA:55 HEXB:94 NPC1:125 NAGLU:131 SGCG:155 RPS27:253
Pancreatitis	−0.10378062	165	4.517e−06	1.926e−03	PNLIP:8 ALB:9 CTRC:13 CYP1A1:41 PNLIPRP2:46 CELA3B:117
Kidney Calculi	−0.18184705	53	4.746e−06	1.939e−03	CLDN16:1 SLC4A1:17 AGXT:39 SLC13A2:50 SLC2A9:202 CAT:240
Hepatitis, Drug−Induced	−0.10327933	164	5.359e−06	2.102e−03	ALB:9 IL1R2:49 CA3:71 NR1I2:158 SULT1E1:170 CAT:240
Atherogenesis	−0.19831912	43	6.901e−06	2.603e−03	NPC1:125 ABCG5:296 TRPV1:309 HRH1:356 CNR2:524 PTGS2:578
Metabolic Diseases	−0.06990403	349	8.285e−06	3.009e−03	GCDH:10 TAT:24 AGXT:39 ACY1:83 SCD:92 HGD:124
Hyperlipidemia	−0.09597714	181	9.034e−06	3.163e−03	ALB:9 MTTP:123 NR1I2:158 PLA2G1B:162 CYP7A1:220 CHDH:241
Hereditary spherocytosis	−0.26604779	23	1.008e−05	3.407e−03	ANK1:16 SLC4A1:17 SDS:47 SPTA1:48 EPB42:214 SLC26A8:257
Tetany	−0.28319069	20	1.169e−05	3.820e−03	CLDN16:1 KCNJ1:37 HIRA:68 COMT:74 TRPM6:78 SLC12A3:249
Coronary Artery Disease	−0.04629911	793	1.270e−05	4.017e−03	ALB:9 CYP1A1:41 PROM1:45 CYP17A1:63 FGD5:70 COMT:74
Hyperkalemia	−0.30109555	17	1.729e−05	5.298e−03	CYP11A1:22 CYP17A1:63 WNK4:197 SCNN1A:212 SCNN1B:217 CA12:666
Malabsorption Syndrome	−0.21132381	34	2.023e−05	6.011e−03	PNLIP:8 SLC10A2:30 FABP1:72 LCT:86 MTTP:123 CUBN:161
Dyslipidemias	−0.09172485	181	2.211e−05	6.377e−03	ALB:9 SCD:92 GOT2:93 MTTP:123 RORA:140 NR1I2:158
Peripheral Vascular Diseases	−0.14229186	72	3.048e−05	8.540e−03	ALB:9 AGXT:39 CYP1A1:41 MTTP:123 ABCG5:296 ABCG8:406
Hypercalciuria	−0.18549700	41	4.002e−05	1.090e−02	CLDN16:1 KCNJ1:37 WNK4:197 SLC12A3:249 VDR:305 PTH:337
Nonalcoholic Steatohepatitis	−0.09106562	168	4.906e−05	1.300e−02	CYP17A1:63 MTTP:123 CPT1A:139 LAMP1:350 DECR1:380 SIRT6:428
Diabetic Retinopathy	−0.08224007	203	5.709e−05	1.386e−02	ALB:9 SDS:47 PGF:154 SARDH:213 CAT:240 SLC12A3:249
Hypocalcemia	−0.17778733	43	5.550e−05	1.386e−02	CLDN16:1 SLC4A1:17 HIRA:68 COMT:74 TRPM6:78 CHDH:241
Hypochloremia (disorder)	−0.47290232	6	6.032e−05	1.386e−02	KCNJ1:37 CLCNKA:108 CLCNKB:134 SLC26A3:203 SLC12A1:867 BSND:1023

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0003635 abnormal synaptic transmissio	0.10470794	390	6.014e−12	2.857e−09	CAMK2B:4 SNAP25:11 ADCY1:37 EPHA4:41 GRIK3:102 SYT1:113
MP0005332 abnormal amino acid	−0.17812792	113	8.893e−11	1.408e−08	GNMT:29 SLC6A6:40 ATG7:90 BCAT2:114 HGD:124 ABCC6:153
MP0009642 abnormal blood homeostasis	−0.06256728	1196	5.990e−11	1.408e−08	CLDN16:1 SLC23A1:6 PNLIP:8 ALB:9 CLPS:12 SLC4A1:17
MP0009643 abnormal urine homeostasis	−0.12548896	218	3.447e−10	4.094e−08	CLDN16:1 SLC23A1:6 ALB:9 GCDH:10 SLC14A2:14 SLC4A1:17
MP0000609 abnormal liver physiology	−0.08878397	360	1.884e−08	1.790e−06	EHHADH:5 ALB:9 SLC10A2:30 MAN2A1:34 CYP1A1:41 FABP1:72
MP0005636 abnormal mineral homeostasis	−0.14329072	129	2.630e−08	2.082e−06	CLDN16:1 ALB:9 ANK1:16 SLC4A1:17 CYP11A1:22 KCNJ1:37
MP0002118 abnormal lipid homeostasis	−0.08495170	380	3.483e−08	2.363e−06	EHHADH:5 PNLIP:8 ALB:9 CLPS:12 CYP11A1:22 GNMT:29
MP0000598 abnormal liver morphology	−0.07192665	443	5.510e−07	3.272e−05	PNLIP:8 ANK1:16 GNMT:29 MAN2A1:34 CYP1A1:41 MANBA:55
MP0002136 abnormal kidney physiology	−0.08750026	267	1.441e−06	7.607e−05	CLDN16:1 SLC23A1:6 ALB:9 SLC14A2:14 SLC4A1:17 MAN2A1:34
MP0008469 abnormal protein level	−0.07046566	373	5.737e−06	2.725e−04	GNMT:29 CYP1A1:41 ATG7:90 HEXB:94 MTTP:123 SGCG:155
MP0005319 abnormal enzyme/ coenzyme	−0.09453140	166	3.346e−05	1.397e−03	GNMT:29 CYP1A1:41 SLC1A1:60 ATG7:90 SGCG:155 NR1I2:158
MP0005464 abnormal platelet physiology	−0.13714191	77	3.529e−05	1.397e−03	SULT1E1:170 F2RL2:293 ABCG5:296 BAK1:362 FGG:395 VASP:563
MP0001765 abnormal ion homeostasis	−0.11489438	99	8.834e−05	3.228e−03	CLDN16:1 SLC4A1:17 CYP11A1:22 KCNJ1:37 TRPM6:78 RHCG:157
MP0003656 abnormal erythrocyte physiolo	−0.23555847	21	1.906e−04	6.466e−03	ANK1:16 SLC4A1:17 MAN2A1:34 ABCG5:296 AQP9:397 KCNN4:1069
MP0002078 abnormal glucose homeostasis	−0.05293015	427	2.898e−04	8.725e−03	PNLIP:8 GNMT:29 SLC6A6:40 FABP1:72 ABCB11:82 BCAT2:114
MP0002063 abnormal learning/memory/cond	0.05960975	329	2.939e−04	8.725e−03	SNAP25:11 ADCY1:37 MID1:119 HOMER2:133 MBP:157 NRCAM:173
MP0005085 abnormal gallbladder physiolo	−0.21239269	24	3.238e−04	9.048e−03	SLC10A2:30 ABCB11:82 CYP7A1:220 ATP8B1:229 ABCG5:296 ABCG8:406
MP0004019 abnormal vitamin homeostasis	−0.16027595	41	3.998e−04	1.055e−02	CLDN16:1 SLC23A1:6 CUBN:161 CYP7A1:220 VDR:305 LRAT:636
MP0001845 abnormal inflammatory respons	−0.04814115	475	5.432e−04	1.122e−02	MAN2A1:34 SLC9A4:99 PLAUR:196 SLC2A9:202 SLC7A9:246 IKBKB:275
MP0005551 abnormal eye electrophysiolog	0.08736892	136	4.992e−04	1.122e−02	CPLX3:5 EGFLAM:35 GABRR1:56 NPTX1:57 GNAT2:78 ACTA2:85
MP0002081 perinatal lethality	0.04012002	742	4.736e−04	1.122e−02	SNAP25:11 FGF8:46 ITGB4:91 NBEA:125 SYNE1:162 KIF7:175
MP0002272 abnormal nervous system	0.07706758	175	5.216e−04	1.122e−02	GABRR1:56 PLEC:83 GRIK1:165 NRCAM:173 CHRM2:196 GRIK2:249
MP0005391 vision/eye phenotype	0.07592017	181	5.134e−04	1.122e−02	SRSF1:22 EGFLAM:35 PFDN5:93 TRPM1:111 DMD:174 ATP1B2:211
MP0002882 abnormal neuron morphology	0.03761069	813	6.706e−04	1.327e−02	CPLX3:5 MAGOH:21 SRSF1:22 EGFLAM:35 EPHA4:41 CRYBB2:43
MP0001663 abnormal digestive system	−0.10796154	83	7.284e−04	1.352e−02	SLC9A4:99 GLG1:120 SLC26A3:203 SCNN1B:217 OLFM4:291 SLC22A1:379
MP0001666 abnormal nutrient absorption	−0.15464651	40	7.400e−04	1.352e−02	SLC23A1:6 PNLIP:8 ALB:9 CLPS:12 SLC10A2:30 PNLIPRP2:46
MP0005193 abnormal anterior eye	0.06579220	225	8.346e−04	1.468e−02	CRYAA:1 BNC1:2 MAB21L2:12 HSF4:26 CRYBA1:36 CRYBB2:43
MP0002752 abnormal somatic nervous	0.04387992	533	9.068e−04	1.538e−02	CPLX3:5 SRSF1:22 EGFLAM:35 EPHA4:41 FGF8:46 MOG:71
MP0005253 abnormal eye physiology	0.10153040	89	1.008e−03	1.651e−02	SRSF1:22 EGFLAM:35 PFDN5:93 TRPM1:111 DMD:174 ATP1B2:211
MP0003861 abnormal nervous system	0.03996049	642	1.054e−03	1.668e−02	MAGOH:21 EPHA4:41 FGF8:46 ICMT:50 DCLK1:55 CDC42:77
MP0005381 digestive/alimentary phenotyp	−0.09297890	105	1.093e−03	1.674e−02	SLC9A4:99 GLG1:120 SLC26A3:203 SCNN1B:217 OLFM4:291 MYH11:365
MP0002152 abnormal brain morphology	0.03431855	873	1.407e−03	2.089e−02	SNAP25:11 MAGOH:21 ADCY1:37 EPHA4:41 CRYBB2:43 FGF8:46
MP0004043 abnormal pH regulation	−0.24448524	14	1.555e−03	2.222e−02	KCNJ1:37 RHCG:157 WNK4:197 SCNN1B:217 SLC12A1:867 SLC4A7:1189
MP0000751 myopathy	0.20953960	19	1.590e−03	2.222e−02	PLEC:83 DMD:174 COL6A1:532 LAMA2:734 MTM1:924 COL13A1:1296
MP0000230 abnormal systemic arterial	−0.07717055	139	1.886e−03	2.478e−02	RORA:140 DRD3:195 WNK4:197 SCNN1B:217 SLC12A3:249 VDR:305
MP0003868 abnormal feces composition	−0.17948882	25	1.930e−03	2.478e−02	PNLIP:8 ALB:9 CLPS:12 SLC10A2:30 SLC26A3:203 CYP7A1:220
MP0000759 abnormal skeletal muscle	0.06043012	232	1.847e−03	2.478e−02	SNAP25:11 PLEC:83 HSPB2:120 DOCK1:132 DNAJC5:151 SYNE1:162
MP0005365 abnormal bile salt	−0.17114501	26	2.571e−03	3.170e−02	EHHADH:5 PNLIP:8 ALB:9 SLC10A2:30 ABCB11:82 CYP7A1:220
MP0000049 abnormal middle ear	0.11894029	54	2.603e−03	3.170e−02	FGF8:46 TSHZ1:316 PRKRA:328 DUSP6:447 PGAP1:617 POU3F4:768
MP0005584 abnormal enzyme/coenzyme acti	−0.07477615	134	3.096e−03	3.677e−02	CACNA1A:159 DRD3:195 CAT:240 IKBKB:275 VDR:305 NQO1:485

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
liver	−0.15750609	322	1.216e−21	6.564e−20	EHHADH:5 ALB:9 TAT:24 PBLD:26 AGXT:39 SDS:47
peyers.patch	−0.14301079	163	4.142e−10	1.118e−08	SLC5A12:3 SLC23A1:6 SLC10A2:30 SLC13A2:50 FABP1:72 MTTP:123
cortex.kidney	−0.16784870	110	1.462e−09	2.631e−08	CLDN16:1 SLC5A12:3 SLC6A12:36 KCNJ1:37 FABP1:72 ENAM:89
Brodmann.area.9	0.12346141	162	7.522e−08	1.015e−06	VSNL1:31 DCLK1:55 SYT1:113 SLITRK1:138 NRCAM:173 GRIN2B:294
cerebellar.hemisphere	0.07427783	459	1.015e−07	1.053e−06	CAMK2B:4 SNAP25:11 ADCY1:37 NPTX1:57 KCND3:69 SLITRK4:70
pancreas	−0.15969751	93	1.170e−07	1.053e−06	CPB1:4 PNLIP:8 CLPS:12 CTRC:13 GNMT:29 PRSS3:35
thyroid	−0.10963160	147	5.253e−06	4.052e−05	HIRA:68 IDO2:91 CLCNKA:108 PGF:154 PRTG:200 IYD:320
cerebral.cortex	0.12428165	103	1.451e−05	8.706e−05	C1QL2:140 ARNT2:206 KCNAB2:247 SLC17A7:318 GRIN1:415 CHRM1:456
transverse.colon	−0.10559125	144	1.408e−05	8.706e−05	FABP1:72 TRPM6:78 CA2:146 NR1I2:158 CA1:160 LGALS4:194
cerebellum	0.06567197	354	3.077e−05	1.662e−04	TRIM46:166 FAM135B:202 ATP1B2:211 FUT9:225 GPR179:230 ZIC4:237
stomach	−0.12137463	84	1.284e−04	6.303e−04	PGC:18 PGA4:19 TMPRSS2:51 PGA5:58 TMED6:64 PGA3:81
hypothalamus	0.13925103	58	2.544e−04	1.145e−03	GRIK1:165 TMEM255A:632 CARTPT:806 DLX2:862 TMEM196:869 DIRAS3:997
skeletal.muscle	0.06828128	232	3.988e−04	1.538e−03	FSD2:29 HSPB3:49 ALPK3:108 CSRP3:123 SMTNL2:220 MYH8:370
transformed.skin.fibroblast	0.06680565	243	3.963e−04	1.538e−03	BNC1:2 ADAMTS12:136 SH3BP4:156 CRIM1:236 TENM3:305 FAT1:332
nucleus.accumbens	0.10094357	93	8.113e−04	2.921e−03	JAKMIP1:32 PRKCG:53 KCTD4:264 DACH2:396 SLC32A1:566 KIRREL3:738
blood	−0.05901043	266	1.082e−03	3.653e−03	SLC4A1:17 TREML2:23 SLC6A6:40 IL1R2:49 GP9:52 RHAG:53
amygdala	0.31842144	8	1.822e−03	5.787e−03	PTPRZ1:295 SLC17A7:318 NEUROD6:875 TNR:1728 NEUROD2:3637 SLC38A8:3884
tibial.artery	0.07337257	101	1.124e−02	3.371e−02	ACTA2:85 C1QTNF1:170 AMIGO2:323 CDH6:572 TAGLN:729 TLN1:833
C1.spinal.cord	0.05471863	171	1.439e−02	4.090e−02	MOG:71 MBP:157 SYNJ2:267 GPM6B:375 MYH15:404 SEMA4D:440
sigmoid.colon	0.09560925	50	1.964e−02	5.050e−02	MAB21L2:12 TACR2:116 ACTG2:130 CHRM2:196 HAND1:226 DES:283
tibial.nerve	0.05075052	183	1.897e−02	5.050e−02	ITGB4:91 PRRG3:99 GRIK3:102 DMD:174 STARD13:195 ITGA6:277
minor.salivary.gland	−0.07767585	69	2.617e−02	6.424e−02	BPIFB1:2 PROM1:45 ENPP3:88 SCNN1A:212 SLC6A14:352 PRSS8:639
pituitary.gland	0.04498198	202	2.914e−02	6.843e−02	KCNH6:241 GAL:252 UNC79:253 NSG1:357 SCN3B:398 AHI1:452
eye.development	0.08038348	57	3.632e−02	8.171e−02	DLX2:862 FGFR2:950 NOTCH2:1105 RAX:1132 HES5:1144 NRL:1167
bladder	−0.11819017	23	4.999e−02	9.628e−02	CLEC3A:346 UGT1A6:1131 UPK1A:1154 PADI3:1220 CYP4B1:1625 UPK1B:1746
caudate.nucleus	0.11037575	26	5.168e−02	9.628e−02	GABRG1:928 PDE1B:1013 PPP1R1B:1345 GPR88:1479 LRRC10B:1582 RLBP1:1631
EBV.lymphocyte	−0.02319063	652	5.107e−02	9.628e−02	TMEM131L:61 CHAC2:66 CRYZ:107 CHRNA6:178 SGO1:183 ACADM:312
left.ventricle	0.06895875	67	5.171e−02	9.628e−02	ACTC1:28 FSD2:29 HSPB3:49 CSRP3:123 CRYAB:268 SMPX:506
testis	−0.01717489	1339	5.092e−02	9.628e−02	CFAP20DC:44 SPTA1:48 GP9:52 CYP2F1:57 TIGD4:77 IDO2:91
adrenal.gland	−0.05353488	107	5.703e−02	1.026e−01	SLC14A2:14 CYP11A1:22 CYP17A1:63 CELA1:264 MC2R:284 ABCB1:335
Brodmann.area.24	0.09522772	29	7.630e−02	1.329e−01	NCS1:48 SLC17A7:318 NEUROD6:875 FAM189A1:1036 CRH:1163 RAPGEF4:1954
esophagus.muscularis.mucosa	0.09896886	26	8.104e−02	1.368e−01	SYNM:488 KCNMB1:618 RGS2:756 CHRM3:867 COL4A6:902 EPHA7:1346
spleen	−0.02932984	291	8.993e−02	1.472e−01	GP9:52 PLA2G2D:278 DENND1C:392 SAP25:408 ADAMDEC1:422 PATL2:435
hippocampus.proper	0.15726095	9	1.024e−01	1.586e−01	SLC17A7:318 NEUROD6:875 NEUROD2:3637 FEZF2:3903 OLIG2:5107 NEUROG3:5578
substantia.nigra	−0.14213036	11	1.028e−01	1.586e−01	SLC6A11:1693 DBX2:1831 SLC6A3:2193 C10orf105:3696 KCNE5:4566 TH:4664
prostate	0.07304452	37	1.248e−01	1.872e−01	HOXB13:168 NR2E3:539 NPY:746 ARHGEF38:1181 SRD5A2:1263 ACP3:1368
endocervix	0.04996622	70	1.497e−01	2.185e−01	ELFN1:374 BDKRB1:381 ZNF516:402 ISLR:477 NXNL2:557 REM1:868
aorta	0.03134156	151	1.871e−01	2.659e−01	GABRR1:56 ADAMTSL3:117 PCDH10:255 CNTN4:359 ECM2:411 PDGFA:429
lung	−0.02981133	139	2.282e−01	3.160e−01	GP9:52 GPRC5A:133 HTR3C:141 ADGRF5:184 MSR1:243 A2M:463
ovary	0.03246258	98	2.691e−01	3.633e−01	H3−2:80 TMEM98:127 TCF23:146 C21orf62:385 MMP21:522 ANGPTL5:645